

Ahmed Akram Amer

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PROFESSIONAL SUMMARY

Final year Computer Science and Engineering student (GPA 3.71/4.0) specializing in LLM applications and multi-agent systems, with 3+ production-ready AI systems deployed using LangGraph, CrewAI, and RAG pipelines. Experienced in building scalable, observable, and reliable AI agents for end-to-end automation — from research pipelines to enterprise-grade procurement systems. Proficient in Python and the modern AI/ML stack, with proven results including up to 95% model accuracy and 80% reduction in manual workflows.

PROJECTS

Agent System — Multi-Agent Research and Writing | *Python, LangGraph, CrewAI, FastAPI, n8n, Groq LLaMA 3.3* Jan 2025 – Mar 2025

- Built a multi-agent pipeline that cut manual research time by ~80% and generated structured reports in under 2 minutes, connecting n8n, FastAPI, LangGraph, and CrewAI agents end-to-end.
- Designed a router node dynamically dispatching tasks to specialized agents: Research Analyst (SerperDev/Google Search) and Content Writer — with full AgentOps observability across all runs.
- Integrated Groq LLaMA 3.3 70B achieving GPT-4 comparable output without incurring API costs; resolved 6+ integration challenges including Poetry cache conflicts and Arabic encoding in n8n.

Personal Agentic Assistant — Full-Stack RAG App | *Python, FastAPI, LangChain, Pinecone, Google Gemini, Streamlit* Nov 2024 – Jan 2025

- Built a full-stack agentic RAG application enabling users to upload PDF documents (50+ pages) and chat with them via a LangChain ReAct agent powered by Gemini 2.5 Flash.
- Backend: FastAPI and Pinecone vector database with LangSmith observability, achieving sub-3s response times. Frontend: Streamlit ChatGPT-like interface with real-time upload and multi-turn memory.
- Implemented CI/CD pipelines with GitHub Actions to automate testing and deployment of the FastAPI backend.

AI-Powered Procurement System — Multi-Agent | *Python, CrewAI, LangChain, Groq LLaMA 3.3, Tavily, AgentOps* Sep 2024 – Nov 2024

- Designed a 4-agent CrewAI pipeline automating the full procurement workflow: query generation, product discovery, web scraping, and Bootstrap HTML report generation, cutting manual research by 70%.
- Containerized the CrewAI agents and orchestrated them with Docker for high-availability processing; agents compare prices across 3 platforms (Amazon Egypt, Jumia, Noon) in real-time.
- Integrated AgentOps for monitoring token usage, cost, fail rate, and execution traces — achieving full observability across all agent runs.

Customer Support Chatbot — Multilingual RAG | *Python, LangChain, Chroma, Groq, HuggingFace, Gradio* Jul 2024 – Sep 2024

- Built a production-style RAG chatbot handling customer queries in Arabic and English from a PDF knowledge base using multilingual embeddings (intfloat/multilingual-e5-small) with MMR retrieval (k=6).
- Implemented hallucination control via strict system prompting — agent responds only from retrieved context or escalates to human support.

Deep Learning with PyTorch — Multi-Domain | *Python, PyTorch, CNNs, Transfer Learning, Audio Classification* Apr 2024 – Jul 2024

- End-to-end PyTorch pipelines for image classification (CIFAR-10, MNIST), audio classification, e-commerce clothing, and breast cancer detection — with regularization and inference-ready deployment.

German Traffic Signs Classification — Computer Vision | *Python, TensorFlow, OpenCV* Feb 2024 – Apr 2024

- Achieved 95% accuracy on the GTSRB dataset using CNN with transfer learning; reduced training time by 25% — applicable to autonomous driving use cases.

ML Lifecycle Factory — ML Forge | *Python, Flask, MLflow* Oct 2023 – Feb 2024

- Architected a web platform automating the full ML lifecycle (ingestion to REST API deployment) as a factory pipeline, reducing overall ML cycle time by 40% via MLflow experiment tracking.

House Price Prediction — Regression | *Python, XGBoost, Scikit-learn* Aug 2023 – Oct 2023

- Developed an XGBoost regression pipeline achieving Train $R^2=0.95$ and Test $R^2=0.85$ on California housing data via hyperparameter tuning and cross-validation.

TECHNICAL SKILLS

AI Agents	LangGraph, CrewAI, LangChain, LangSmith, AgentOps, RAG, ReAct, Prompt Engineering
LLMs & NLP	Hugging Face Transformers, NLTK, SpaCy, Groq, Google Gemini, OpenAI
Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV
Web & APIs	FastAPI, Flask, Streamlit, Gradio
MLOps & Cloud	MLflow, Docker, GitHub Actions (CI/CD), AWS (EC2, S3, SageMaker), Pinecone, Chroma, LangSmith
Data	Pandas, NumPy, Matplotlib, Seaborn, Plotly, Dash
Languages	Python, SQL (SQL Server, Oracle, SQLite)
Tools	Git, GitHub, Jupyter, Google Colab, VS Code

EXPERIENCE

- AI & Data Science Intern • DEPI | EYouth** *Jun 2024 – Dec 2024*
- Engineered supervised ML models for predictive analytics, achieving 90% accuracy on structured datasets and reducing manual reporting time by 50%.
 - Conducted EDA and feature engineering with Pandas and Seaborn across 10+ high-dimensional datasets, surfacing actionable business insights for stakeholders.
 - Deployed supervised ML models as REST APIs using Flask and Docker, enabling real-time inference for business users and reducing inference latency in production environments.

EDUCATION

- Bachelor of Electronic Engineering — Computer Science and Engineering** *Expected Oct 2026*
- Menoufia University • GPA: 3.71 / 4.0

CERTIFICATIONS

- Machine Learning Specialization** — DeepLearning.ai • **AI Engineer for Data Scientists Associate** — DataCamp • **HCIA-AI** — Huawei
- Python for AI and Data Science** — IBM • **Data Scientist** — DEPI and EYouth • **Applied Deep Learning** — MaharaTech • **Sprints x Microsoft AI Camp**